

9.12 Solution: The total volume of the sphere is

$$\begin{aligned} V(\beta) &= \int_0^{2\pi} d\phi \int_{-1}^1 d(\cos \theta) \int_0^{R(\theta)} r^2 dr = \frac{2\pi R_0^3}{3} \int_{-1}^1 [1 + \beta P_2(\cos \theta)]^3 d(\cos \theta) \\ &= \frac{4\pi R_0^3}{3} \left(1 + \frac{3}{5}\beta^2\right) = V_0 \left(1 + \frac{3}{5}\beta^2\right), \end{aligned}$$

where $V_0 = 4\pi R_0^3/3$ is the volume of a perfect sphere. Then, the charge density is

$$\rho(\beta) = \frac{Q}{V(\beta)} = \frac{Q}{V_0} \left(1 + \frac{3}{5}\beta^2\right)^{-1} \approx \rho_0 \left(1 - \frac{3}{5}\beta^2\right) \approx \rho_0,$$

to first order in β , assuming $\beta \ll 1$. Here,

$$\rho_0 = \frac{Q}{4\pi R_0^3/3}.$$

Therefore, we can calculate the multipole moments as if the charge density stays the same, while the shape of the sphere oscillates.

The dipole moment can be calculated as

$$\mathbf{p} = \rho_0 \int \mathbf{r} d^3x = \rho_0 \int_0^{2\pi} d\phi \int_{-1}^1 d(\cos \theta) \int_0^{R(\theta)} r^3 (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta) dr.$$

The p_x and p_y components are clearly zero, and we only need to consider the p_z component,

$$p_z = 2\pi\rho_0 \int_{-1}^1 \cos \theta \cdot \frac{1}{4} (1 + \beta P_2(\cos \theta))^4 d(\cos \theta) = 0.$$

Thus, to the lowest approximation in β , there is no dipole moment.

The quadrupole moments are

$$Q_{\alpha\beta} = \rho_0 \int (3x_\alpha x_\beta - r^2 \delta_{\alpha\beta}) d^3x.$$

It is straightforward to verify that the off-diagonal elements are all zero, and we only need to consider the diagonal ones,

$$\begin{aligned} \begin{pmatrix} Q_{xx} \\ Q_{yy} \\ Q_{zz} \end{pmatrix} &= \rho_0 \int_0^{2\pi} d\phi \int_{-1}^1 d(\cos \theta) \int_0^{R(\theta)} r^4 \begin{pmatrix} 3 \sin^2 \theta \cos^2 \phi - 1 \\ 3 \sin^2 \theta \sin^2 \phi - 1 \\ 3 \cos^2 \theta - 1 \end{pmatrix} dr \\ &= \frac{2\pi\rho_0 R_0^5}{5} \int_{-1}^1 (1 + \beta P_2(\cos \theta))^5 \begin{pmatrix} \frac{1}{2} - \frac{3}{2} \cos^2 \theta \\ \frac{1}{2} - \frac{3}{2} \cos^2 \theta \\ 3 \cos^2 \theta - 1 \end{pmatrix} d(\cos \theta) \\ &= \frac{2\pi\rho_0 R_0^5}{5} \begin{pmatrix} -2\beta \\ -2\beta \\ 4\beta \end{pmatrix} = \frac{3}{5} Q R_0^2 \begin{pmatrix} -\beta \\ -\beta \\ 2\beta \end{pmatrix}. \end{aligned}$$

Using Eq. (9.51), with $Q_0 = (6QR_0^2/5)\beta_0$, for the quadrupole radiation,

$$\frac{dP}{d\Omega} = \frac{c^2 Z_0 k^6}{512\pi^2} \frac{36Q^2 R_0^4}{25} \beta_0^2 \sin^2 \theta \cos^2 \theta = \frac{9c^2 Z_0 k^6 Q^2 R_0^4}{3200\pi^2} \beta_0^2 \sin^2 \theta \cos^2 \theta,$$

and

$$P = \frac{c^2 Z_0 k^6}{960\pi} \frac{36Q^2 R_0^4}{25} \beta_0^2 = \frac{3c^2 Z_0 k^6 Q^2 R_0^4}{2000\pi} \beta_0^2.$$